LLM Loan Approval Application Design Guide - Updated

MTB Query 1, Query 2 and Query 4 | Underwriting eligibility, product rules, risk assessment and explainable decisions

Purpose: Implement the supplied underwriting notes as configurable policy checks ahead of LLM-assisted analysis. Product limits and caveats below reflect the provided screenshot and must be confirmed by lending policy owners before production use.

1. Reference Architecture

Excel/MTB extracts -> schema validation -> entity and loan joins -> normalized borrower/financial/credit profile -> product and Quick App eligibility -> risk rules and exception routing -> LLM explanation -> underwriter review -> auditable decision.

Use deterministic rules for eligibility, limits, OFAC handling, credit freezes, and mandatory documentation. The LLM summarizes evidence and may recommend next steps but cannot change a rule outcome.

2. Dataset Usage and Mapping

Dataset	Primary role	Example fields
Query 1	Master underwriting and business/loan data	Business name/type, annual revenue, net income, EBIT, net worth, assets, liabilities, cash, amount, product, group credit, SBSS7 score
Query 2	Indicators and related borrower/entity records	SCOC__IndicatorKey__c, SCOC__IndicatorValue__c, SBSS7 score, borrower type, loan/account references
Query 4	Credit bureau and alert attributes	Credit score, credit report freeze, past due, bankruptcies, charge-offs, liens, judgments, OFAC, collection, critical alerts

Join by validated account, entity and loan identifiers; resolve borrower versus guarantor roles. Do not join unrelated records by row order. Query 2 key/value indicators should be pivoted per correct borrower and request. Missing values remain unknown, never false.

3. Business Definitions and Eligibility

Concept	Supplied definition / treatment
Quick App	Total new request <= $100M and Tier I Group Credit <= $250M. Not-for-profit (NFP) entities do not qualify; route to full underwriting.
Express	Unsecured line of credit or term loan / BIL at or below $100M.
Total Group Credit (TGC)	Entire lending relationship across entities and guarantors; Tier I is considered TGC unless Tier II group credit is greater. Above $250M: submit through nCino per supplied notes.
Tier I Group Credit	BCC, TL, BIL and BALOC (actual loans).
Tier II Group Credit	Examples include merchant guidance line, cash management ACH and purchasing.

Boundary handling: The screenshot uses <= $100M and <= $250M. Amounts above a boundary are not Quick App eligible. Confirm whether limits are in actual dollars or another unit in the source systems.

4. Product Eligibility and Special Conditions

Product	Rules from supplied underwriting notes
Express BALOC	Unsecured; <= $100M. An increase that puts LOC above $100M may still be processed through Quick App when total new money is below $100M and group credit remains below $250M, subject to policy confirmation.
Conventional BALOC	Above $100M; must be secured.
BIL (Business Installment Loan)	<= $100M unsecured term debt; acceptance letter required. If collateral is taken (e.g., vehicle), change to term loan as applicable.
SBA BIL	Permanent working capital only per notes. Flat fee, no PPP; fixed rates with principal and interest payments. Same SBA process as BALOC, including SBA memo and CAM.
Term Loan	May be secured or unsecured. Apply target pricing spread, COF index, fixed-rate custom-field language, PPP stipulations and collateral where applicable. Notes mention unsecured equipment/vehicle under $20M.

5. BALOC Caveats and Exceptions

• SBA and conventional BALOC cannot be converted into each other in place; create a new line.

• Within conventional BALOC, secured/unsecured and express/standard may be changed; if reducing a secured line below $100M, retain collateral per notes.

• If conventional BALOC scores as SBA, requested line must be doubled for SBA consideration per supplied guidance; obtain business policy validation before automating.

• Line increases are not permitted within six months of account opening; route to policy exception handling rather than silently approving.

• Investment real-estate company BALOC cap: $25M. Startup BALOC cap: $25M revolving; supplied notes say term loans do not share this limitation.

6. Recommended Deterministic Decision Rules

Rule / condition	Proposed system action	Notes
Confirmed OFAC match	BLOCK / compliance escalation	Prior guide says DENY; do not treat an unverified alert as a confirmed match. Compliance must validate disposition.
Credit report freeze	MANUAL REVIEW / hold	No automatic credit approval until required bureau information is available.
NFP requesting Quick App	FULL UNDERWRITING	Not a loan denial.
Quick App size/TGC exceeds limit	FULL UNDERWRITING or nCino routing	TGC > $250M follows supplied nCino instruction.
Product or BALOC caveat fails	INELIGIBLE FOR REQUESTED PATH / REVIEW	Provide specific rule and remediation; not automatically deny all financing.
Multiple bankruptcies	HIGH RISK REVIEW	Original guide says DENY/REVIEW; number, recency and approved policy threshold are unspecified.

Rule / condition	Proposed system action	Notes
High past-due balances	MANUAL REVIEW	Threshold and ratio are not specified.
Strong SBSS and financials	APPROVAL CANDIDATE	Requires all other checks and human approval.

Policy parameters to configure: threshold for multiple bankruptcies, past-due severity, minimum SBSS, acceptable credit score, debt-service/coverage measures, financial ratios, recent adverse credit, and exception approval authorities. No thresholds should be invented by the LLM.

7. Decision State Model

Use internal statuses: ELIGIBLE_FOR_SCORING, FULL_UNDERWRITING, MANUAL_REVIEW, COMPLIANCE_HOLD, PRODUCT_INELIGIBLE, APPROVAL_CANDIDATE, DENIAL_RECOMMENDED. Final APPROVE/DENY requires authorized lender workflow, with adverse-action reason controls where applicable. A hold or routing decision is not equivalent to a denial.

8. Feature Engineering and Rule Sequence

• Normalize currency, dates, booleans and missing values; detect duplicates, stale bureau records and borrower/guarantor mismatches.

• Calculate new money, Tier I, Tier II and effective TGC; classify NFP, startup and investment RE; map product family and security/collateral status.

• Apply compliance holds, Quick App and product eligibility, account age, collateral/document requirements, and credit-risk rules in that order.

• Compute transparent financial features such as leverage, liquidity, debt coverage and past-due exposure only when input fields and approved formulas exist.

• Preserve rule IDs, evaluated input values, policy version, evidence source and timestamps.

9. LLM Input Contract (Illustrative JSON)

{ "borrower_profile": {...}, "financial_profile": {...}, "credit_profile": {...}, "indicators": {...}, "loan_request": {"product": "BALOC", "requested_amount": null, "new_money": null}, "group_credit": {"tier_i": null, "tier_ii": null, "effective_tgc": null}, "policy_evaluation": {"rule_results": [], "route": "MANUAL_REVIEW", "policy_version": "TBD"}, "evidence": [] }

10. Expected Structured Output (Illustrative JSON)

{ "recommended_route": "MANUAL_REVIEW", "decision_recommendation": null, "confidence": null, "positive_factors": [], "risk_factors": [], "triggered_rules": [], "missing_information": [], "explanation": "...", "requires_human_approval": true }

The model must cite field-level evidence, avoid inventing values, use the policy route as a fixed input, and return schema-valid JSON. Confidence is an evidence-calibrated model signal, not a probability of repayment or approval.

11. Controls, Testing and Release Criteria

• Human-in-the-loop signoff, OFAC compliance escalation, fair-lending review, model risk governance, access control, PII protection, audit logs and policy versioning.

• Test boundary cases ($100M/$250M), NFP, secured vs unsecured BALOC, six-month line increases, startup and investment RE caps, credit freeze and contradictory source records.

• Measure rule accuracy, missing-data routing, unsupported claims, explanation faithfulness, manual override rates and adverse-action reason consistency.

12. Open Questions Requiring Business Confirmation

• Does TGC equal max(Tier I, Tier II) for every product and entity relationship, and how are guarantor exposures aggregated?

• Is Express BALOC increase eligibility strictly new money < $100M while post-increase LOC can exceed $100M?

• Are BIL and SBA BIL thresholds identical, and what are the exact collateral and SBA eligibility requirements?

• What are the approved SBSS, bankruptcy, past-due, credit-score and financial-ratio thresholds?

• What is the exact disposition for OFAC alerts, product ineligibility and policy exceptions?

Source note: Revised from the uploaded 2-page design guide and the supplied underwriting definitions/products screenshot. This document is an application design proposal, not an approved credit policy.